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Sleep bruxism and its associations with insomnia and OSA in the general population of Sao Paulo

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Abstract

Sleep bruxism (SB) is characterized by recurrent masticatory muscle activity during sleep with occasional tooth grinding. SB can be concomitant with sleep apnea although its association with insomnia is understudied.

Study Objective: Assess the strength of the associations between SB, insomnia and sleep apnea in a general population.

Methods: Data from the 2007 EPISONO general population study (n=1,042; Sao Paulo, Brazil) were reused for the present analyses. The data was collected from polysomnography (PSG) and from a questionnaire. SB could only be assessed as "possible" with self-report questionnaires, but as "definitive" with both self-reports and PSG. Logistic regression and decision tree analyses were performed.

Results: Logistic regression analyses revealed that being male, overweight, obese, having an apnea/hypopnea index (AHI) above 30 and insomnia syndrome are among risk factors for SB (prevalence ratio (PR): 1.5–3.3). A high AHI and insomnia syndrome had similar PRs, 2.7 and 2.8, respectively. Decision tree analysis showed that insomnia syndrome contributed to the predictive accuracy of SB self-report (88%). A similar estimate (91%) was observed with SB PSG data. Correspondence analysis illustrated three age profiles in participants: (1) good sleepers aged 20–35 years, (2) females aged 35–50 years with SB and concomitant insomnia syndrome, and (3) participants aged ≥ 50 years with obesity and sleep apnea.

Conclusions: Insomnia is likely a condition associated with SB, especially in middle-age females, while sleep apnea seems age and gender dependent. Such overlap may influence the treatment decision to achieve best outcomes.

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Key words: Sleep bruxism, insomnia, sleep apnea, obesity, age, gender, polysomnography, correspondence analysis, HOMALS, logistic regression, CHAID, decision tree analysis.

Clinical trial registration: EPISONO study; Clinical trials.gov ID # NCT00596713

Highlights

- Insomnia seems to be associated with sleep bruxism in general population.
- The association between insomnia and sleep bruxism is stronger in middle-aged women.
- Sleep bruxism and sleep apnea associations seem to be age and gender dependent.

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1. Introduction

Sleep bruxism (SB) has been defined as a "masticatory muscle activity during sleep that is characterized as rhythmic (phasic) or non-rhythmic (tonic) and is not a movement disorder or a sleep disorder in otherwise healthy individuals" [1], characterized by recurrent and transient masticatory muscle activity during sleep with occasional tooth grinding [2,3]. Bruxism can be present during wakefulness or sleep, and despite their overlap in some individuals they may have different causes and mechanisms [4,5]. In healthy individuals, SB is not considered a disorder although it can be associated with sleep disorders such as insomnia, sleep-disordered breathing, periodic limb movement, gastroesophageal reflux, and in rare cases, epilepsy or rapid eye movement (REM) behavior disorder (RBD) [1,2]. For the sake of applicability, it was recently proposed that in the absence of critical oral damage or comorbidities, bruxism while awake or asleep should be termed normo-bruxism, and patho-bruxism when accompanied by oral damage or comorbidities (Svensson and Lavigne, in press 2020). The consequences of SB include dental attrition, dental sensitivity, possible fractures of teeth and restorations, and loss of facial vertical dimension. Moreover, SB is also associated with the presence of morning headaches and temporomandibular disorders (TMD) [6]. An international consensus suggested that questionnaires alone can only contribute to a “possible” diagnosis of SB, but when used with clinical examination and polysomnographic recordings (PSG) a more “definitive” diagnosis of SB can be established [3].

In clinical settings, the diagnosis of SB is mostly based on reports of tooth clenching or tooth grinding and transient morning jaw muscle pain or headache, in addition to visual assessment of tooth damage/wear by the clinician. Nevertheless, in research, the exclusive use of questionnaires is unreliable since other oral behaviors

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may overlap. For example, concomitant orofacial pain and mood disorders such as anxiety or depression and associated symptoms can bias the precision of the response and, in addition, not all individuals may be aware of tooth grinding [2,7,8]. Furthermore, tooth wear is not a robust outcome in the clinical assessment of SB, as it may have occurred years ago, and SB episodes may not be exclusively related to tooth contact. Moreover, it may not be related to the frequency of jaw muscle activity during sleep, and its specificity is challenged by other concomitant sleep disorders such as gastric reflux, apnea, and hyposalivation [9-12]. PSG is an aid to confirm the presence of SB by quantification of rhythmic masticatory muscle activity (RMMA) per hour of sleep. While a frequency of RMMA ≥ 4/hour of sleep was initially proposed for research on SB mechanisms, currently a cut-off of ≥ 2/hour of sleep is used in clinical research studies [13-18]. Responses to questionnaires used to assess the awareness of grinding, indicating its occurrence at least four nights a week and the presence of tooth wear, can increase the diagnosis agreement of PSG SB recordings [19], although night to night variability also has to be considered [20,21]. In research, self-report questionnaires are the tools most frequently used to identify “possible” SB, and when used with PSG recording a more “definitive” diagnosis can be obtained [3]. Moreover, PSG can help to exclude some of the comorbidities listed above [1-3]. Nonetheless, the level of uncertainty associated to SB assessment methods and their poor correspondence with clinical outcomes deserve future investigation [22].

In recent years, SB has been associated with sleep disorders such as obstructive sleep apnea (OSA), and also with insomnia. However, this association is a controversial topic in the literature [2,18,23-30]. Survey based research has shown that OSA can be associated with SB, usually in over middle-aged individuals considering that the prevalence of OSA rises with age [31-33]. Studies based on PSG have shown that 54%

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of mild, and 40% of moderate OSA patients had concomitant SB [27], and that 48% of middle-aged OSA participants were diagnosed with SB (OR of 3.9) [34]. In addition, 50% of suspected OSA participants also had SB-related muscle activity ≥ 2 RMMA per hour [18]. Nevertheless, the mechanisms through which SB affects OSA, and vice versa, are still unclear.

The relationship between SB and insomnia also remains unexplored. A noteworthy finding of the EPISONO study was a significant association between SB and insomnia; insomnia was twice as frequent in individuals with SB compared to those without SB [35]. A survey of broadcasting media shift workers revealed that sleep disruption and insomnia symptoms were associated with orofacial pain, bruxism (awake or asleep), gender (female) and age [36,37]. Furthermore, a Canadian survey of a general population showed that almost half of self-reported SB participants (ie, were aware of jaw clenching or grinding during sleep) also complained of sleep maintenance problems [38]. However, despite this initial evidence linking SB and insomnia in both self-report and PSG studies, the relationship and its possible importance is not yet fully elucidated. Although it can be speculated that insomnia and poor sleep quality play some role in bruxism, the influence of risk factors such as age, gender, and comorbidity presence (eg, sleep apnea) need to be further assessed before it can be concluded that there is a clinically relevant association between SB and insomnia. Therefore, in this retrospective study based on data collected from a general population sleep study [35], our main aims were to:

(1) Assess with logistic regression the associations between SB and comorbid insomnia, and SB and sleep apnea, taking into consideration the influence of the following co-variables: age, gender, body mass index (BMI), insomnia symptoms and apnea-hypopnea severity index.

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(2) Identify, using decision tree analysis, the variables that can contribute to improving the predictive value of SB self-reports in a general population.

(3) Use correspondence analysis to illustrate if some profiles emerge when controlling the presence of age, gender, BMI, insomnia symptoms and apnea-hypopnea severity index.

2. Methods

Data were collected in 2007 under the EPISONO study of a general representative population (aged 20–80 years) of Sao Paolo, Brazil, and SB data were published in 2013 [35]. In summary, an independent firm identified the population and demographic data were collected for 1042 participants using a probabilistic three-stage cluster sampling technique in permanently occupied private homes. Slums and shantytowns were excluded due to high criminal activity, and pregnant and lactating women, people with physical or mental impairments that prevent self-care, individuals below 20 or over 80 years old, and people who work every night were not considered. More details about the sampling and recruitment procedures can be found in other publications [39]. All participants agreed to spend one night in a sleep laboratory for PSG data collection under medical supervision. The EPISONO study was registered with the clinical trial registry: Clinical trials.gov ID # NCT00596713 and was approved by the Ethics Committee of the *Universidade Federal de São Paulo* (CAAE: 01570712.4.0000.5505).

Questionnaires, PSG recording methodology and scoring methods are described in detail elsewhere [35]. The presence of SB was categorized in two different ways:

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"possible" SB by self-report, and "definitive" SB by self-report and PSG [1,3]. The categorization of "possible" SB was based on the participant's responses to the questionnaire in respect of their awareness of the following during sleep: tooth grinding, clenching, tooth tapping, or tooth noise on a weekly basis. In this epidemiological study no clinical rating of tooth wear was done. Therefore, it is to be noted that despite clinical ratings were not performed, in this study the term "definitive" was used for clarity purposes, basically indicating more certainty on the diagnosis (ie, confirmed by PSG). Hence, two groups were made:

(1) "Possible" SB by self-report (n=127), which was also compared with a reference group without awareness of SB (i.e., negative SB by self-report (n=915])). The total sample was 1042.

(2) "Definitive" SB cases (n=56), comprising participants who responded positively to the questionnaire in respect of SB symptoms, and had ≥ 2 RMMA episodes per hour in PSG recordings [13,18,35]. For both the PSG logistic regression and the decision tree analyses, we extracted an age (±5 years) and gender matched reference group without self-reported awareness of SB or PSG positive RMMA findings (n=564). The total sample was 620.

The demographic and other questionnaire variables collected were gender (male, female), age (20–29, 30–39, 40–49, 50–59, and 60–80 years), BMI group (eutrophic: BMI < 25, overweight: BMI 25–29.9, and obese: BMI ≥30), and sleep quality using validated questionnaires based on the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition (DSM-IV) criteria [40]. Then, as described previously in a paper from our group, participants were categorized into three groups: good sleepers, insomnia symptoms, and insomnia syndrome [41]. Individuals without insomnia complaints were considered as good sleepers. Individuals classified as insomnia

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symptoms reported difficulties initiating sleep or sleep onset latency greater than 30 minutes, difficulty maintaining sleep and/or early morning awakenings, occurring at least 3 times per week and had little or no daily consequences. The insomnia syndrome group comprised individuals reporting frequent and persistent insomnia symptoms (lasting more than one month) that interfered “much” or “extremely” with daily functioning [30].

Apnea/hypopnea events were recorded and scored according to AASM 2007 criteria, as reported in previous EPISON0 publications [35,42,43]. The apnea and hypopnea index (AHI) was classified as events per hour: normal from 0 to 5, mild from 5 to 15, moderate from 15 to 30, and severe if greater than 30.

The following analyses were done:

(1) Logistic regression with prevalence ratio (PR) to determine which variables were associated with SB using the data obtained from the questionnaires and PSG. The variables age, gender, BMI, sleep quality and insomnia (symptoms and syndrome), and apnea-hypopnea severity index were assessed in both SB groups, “possible” SB (n=127) and “definitive” SB (n=56) [13].

(2) Tree analysis to detect which variables can better contribute to SB recognition using the chi-squared automatic interaction detector (CHAID) algorithm based on questionnaires for all participants (total n=1,042) and PSG (total n=620 [56 SB + 564 non-SB]).

(3) Correspondence analysis, using homogeneity analysis (HOMALS), to classify the distribution of the profiles of the significant SB variables (based on the questionnaire and PSG data) identified from the above logistic regression and CHAID analyses, to visualize the relationships between the multivariate profiles and SB participants.

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For all statistical procedures, an alpha error of 5% ($p<0.05$) was used. The sampling process for the EPISONO study is described in Santos-Silva [39].

3. Results

The descriptive data for the full sample obtained from “possible” SB cases, and “definitive” SB cases is presented in Table 1. For those classified as having insomnia symptoms or insomnia syndrome, the percentage of “possible” SB when compared to “definitive” SB dropped from 12.2% to 8.4% and from 21.3% to 17.7%, respectively. The association of SB with gender distribution and weight remained relatively stable. A steady drop with age seen in both the questionnaire and PSG results, apart from a small rise in the 60–80 group.

Table 1 – Descriptive statistics for the whole sample and sleep bruxism (SB) groups of “possible” SB and “definitive” SB according to questionnaire (A) and PSG criteria (B).

A Questionnaire	**No SB**				**“possible” SB**			
	N	%	95.0% CI		N	%	95.0% CI	
Good sleepers	354	91.20%	88.10%	93.70%	33	8.80%	6.30%	11.90%
Insomnia symptoms	439	87.80%	84.70%	90.40%	61	12.20%	9.60%	15.30%
Insomnia syndrome	122	78.70%	71.80%	84.60%	33	21.30%	15.40%	28.20%
Men	420	89.70%	86.80%	92.20%	47	10.30%	7.80%	13.20%
Women	495	86.10%	83.10%	88.70%	80	13.90%	11.30%	16.90%
20–29 years	202	85.60%	80.70%	89.60%	34	14.40%	10.40%	19.30%
30–39	217	87.50%	83.00%	91.20%	31	12.50%	8.80%	17.00%
40–49	227	89.00%	84.70%	92.40%	27	11.00%	7.60%	15.30%
50–59	149	89.80%	84.50%	93.70%	17	10.20%	6.30%	15.50%
60–80	120	87.00%	80.60%	91.80%	18	13.00%	8.20%	19.40%
Eutrophic	354	84.90%	81.20%	88.10%	63	15.10%	11.90%	18.80%
Overweight	351	87.50%	84.00%	90.50%	50	12.50%	9.50%	16.00%
Obese	209	93.70%	90.00%	96.40%	14	6.30%	3.60%	10.00%
PSG classification AHI < 5	568	87.90%	85.20%	90.30%	78	12.10%	9.70%	14.80%
5 < AHI ≥ 15	193	87.30%	82.50%	91.20%	28	12.70%	8.80%	17.50%
15 < AHI ≤ 30	92	89.30%	82.30%	94.20%	11	10.70%	5.80%	17.70%
AHI ≥ 30	62	86.10%	76.80%	92.60%	10	13.90%	7.40%	23.20%
B PSG	**No SB**				**“definitive” SB**			
	N	%	95.0% CI		N	%	95.0% CI	

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Good sleepers	228	93.10%	89.40%	95.70%	17	6.90%	4.30%	10.60%
Insomnia symptoms	271	91.60%	88.00%	94.30%	25	8.40%	5.70%	12.00%
Insomnia syndrome	65	82.30%	72.80%	89.50%	14	17.70%	10.50%	27.20%
Men	247	91.80%	88.10%	94.70%	22	8.20%	5.30%	11.90%
Women	317	90.30%	86.90%	93.10%	34	9.70%	6.90%	13.10%
20–29	127	88.80%	82.90%	93.20%	16	11.20%	6.80%	17.10%
30–39	128	90.10%	84.40%	94.20%	14	9.90%	5.80%	15.60%
40–49	131	91.60%	86.20%	95.30%	12	8.40%	4.70%	13.80%
50–59	106	93.80%	88.20%	97.20%	7	6.20%	2.80%	11.80%
60–80	72	91.10%	83.40%	95.90%	7	8.90%	4.10%	16.60%
Eutrophic	212	88.70%	84.20%	92.20%	27	11.30%	7.80%	15.80%
Overweight	206	90.40%	86.00%	93.70%	22	9.60%	6.30%	14.00%
Obese	131	94.90%	90.30%	97.70%	7	5.10%	2.30%	9.70%
AHI < 5	336	90.80%	87.50%	93.40%	34	9.20%	6.60%	12.50%
AHI 5–15	125	91.20%	85.60%	95.10%	12	8.80%	4.90%	14.40%
AHI 15–30	54	88.50%	78.80%	94.70%	7	11.50%	5.30%	21.20%
AHI ≥ 30	34	91.90%	79.90%	97.70%	3	8.10%	2.30%	20.10%

Questionnaire: “Possible” SB if positive response for weekly sleep-related tooth grinding, clenching, tapping, or noise.

PSG: “Definitive” SB if presence of ≥ 2 RMMA episodes per hour.

RMMA: Rhythmic Masticatory Muscle Activity; AHI: Apnea-Hypopnea Index.

The logistic regression analysis identified the most significant risk factors associated with “possible” SB and “definitive” SB (Table 2). For “possible” SB, the risk of SB was higher in males (prevalence rate [PR] = 1.56), overweight and obese participants (PR 3.35 & 2.67, respectively), in those with insomnia syndrome (PR 2.68), and in participants with an AHI of 30 and over (PR 2.81). Analysis of the PSG data indicated that in “definitive” SB participants, the only SB risk factors that remained significant were overweight and insomnia syndrome (with a PR close to 3 for each condition; see Table 2B).

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Table 2 – Logistic regression analysis of SB groups of “possible” SB and “definitive” SB according to questionnaire (A) and PSG criteria (B).

A – “Possible” SB (n=127) vs. reference group (n=915)	B	p	Adjusted PR	Adjusted PR, 95% C.I.	
				Lower	Upper
Gender (Male)	**0.45**	**0.03**	**1.56**	**1.03**	**2.36**
Age	0.02	0.06	0.98	0.97	1.00
Eutrophic	-	0.00	-	-	-
Overweight	**1.21**	**0.00**	**3.35**	**1.74**	**6.45**
Obese	**0.98**	**0.00**	**2.67**	**1.41**	**5.07**
Good Sleeper	-	0.00	-	-	-
Insomnia Symptoms	0.42	0.07	1.53	0.97	2.40
Insomnia Syndrome	**0.99**	**0.00**	**2.68**	**1.57**	**4.59**
From PSG classification AHI <5	-	0.07	-	-	-
AHI 5–15	0.47	0.08	1.60	0.94	2.71
AHI 15–30	0.52	0.18	1.68	0.79	3.57
AHI +30	**1.03**	**0.02**	**2.81**	**1.22**	**6.44**

B – “Definitive” SB (n=56) vs. reference group (n=564)	B	p	Adjusted PR	Adjusted PR, 95% C.I.	
				Lower	Upper
Gender (Male)	0.31	0.32	1.37	0.74	2.52
Age	0.02	0.12	0.98	0.96	1.01
Eutrophic	-	0.08	-	-	-
Overweight	**1.09**	**0.02**	**2.98**	**1.16**	**7.68**
Obese	0.88	0.06	2.42	0.97	6.04
Good Sleepers	-	0.02	-	-	-
Insomnia Symptoms	0.31	0.36	1.36	0.70	2.65
Insomnia Syndrome	**1.12**	**0.01**	**3.05**	**1.39**	**6.68**
AHI <5	-	0.36	-	-	-
AHI 5–15	0.42	0.30	1.52	0.69	3.35
AHI 15–30	0.88	0.09	2.40	0.88	6.57
AHI >30	0.71	0.33	2.03	0.49	8.36

Note. **Bold:** significant difference; AHI: apnea-hypopnea index, PR: prevalence ratio; RMMA: rhythmic masticatory muscle activity. The Reference group comprises individuals with absence of sleep bruxism according to both SB questionnaire responses and negative RMMA PSG results. B: Beta Coefficient; p: significant difference ($p<0.05$).

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The CHAID decision tree analysis (Figure 1) revealed 2 models: Model A for “possible” SB and Model B for “definitive” SB cases. The prevalence of SB “possible” cases was 12.3% (n=1042, 127/915; model A), whereas for “definitive” SB cases it was 9% (n=620, 56/564; model B). For Model A, the decision tree analysis correctly classified 21.3% of SB as insomnia cases, compared to 10.6% of SB as good sleepers (p=0.001; Figure 2, Model A). For Model B, 17.7% were classified as insomnia cases compared to 7.8% as SB good sleepers (p=0.008; Figure 1, Model B). Hence, in the presence of insomnia, the prediction of SB being identified using the questionnaire alone or the questionnaire plus the PSG data was 87.7% and 91%, respectively. Furthermore, the decision tree analysis revealed that most participants with SB who are good sleepers have a lower BMI: they are eutrophic or overweight but not obese (12.0% for SB with BMI < 30 vs. 4.6 % for BMI > 30: p=0.027).

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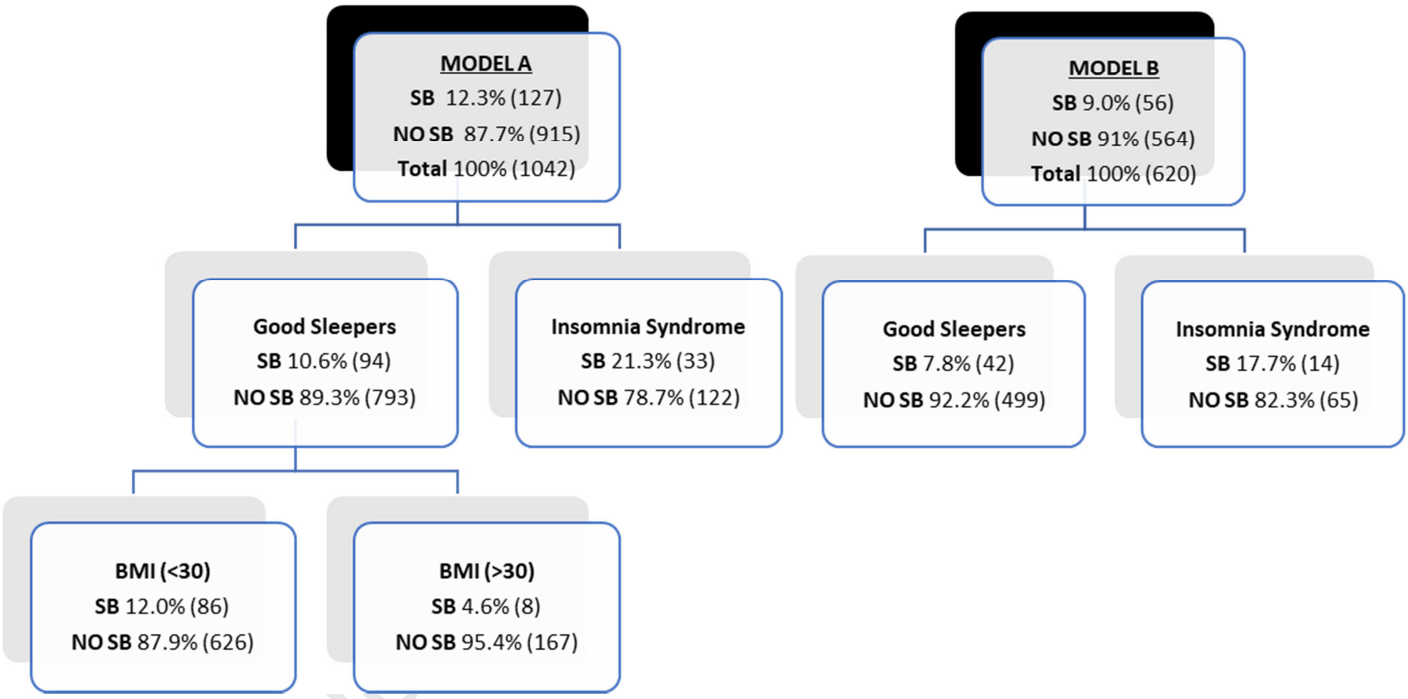
Figure 1 – Decision tree analysis (CHAID Algorithm*) to identify the variables that best contribute to detecting sleep bruxism (age, gender, body mass index [BMI], insomnia syndrome). Model A is based on self-report sleep of bruxism (SB) assessments ("possible" SB), while model B is based on self-report and PSG SB assessments ("definitive" SB), including the AHI.



The correspondence analysis (HOMALS) revealed three profiles of associated variables (Figure 2). First, a group of 20- to 35-year-old participants was characterized by eutrophic weight, good sleep and no sleep apnea. A second group included predominantly females aged 35 to 50 who reported insomnia syndrome and "possible" SB and "definitive" SB. In this age group, men did not present more SB but were overweight. Insomnia symptoms were partly associated with SB females (both self-report and PSG) and non-SB men. A third group without SB, identified by self-report and PSG, contained 50 to 80-year-old obese participants with apnea and hypopnea index

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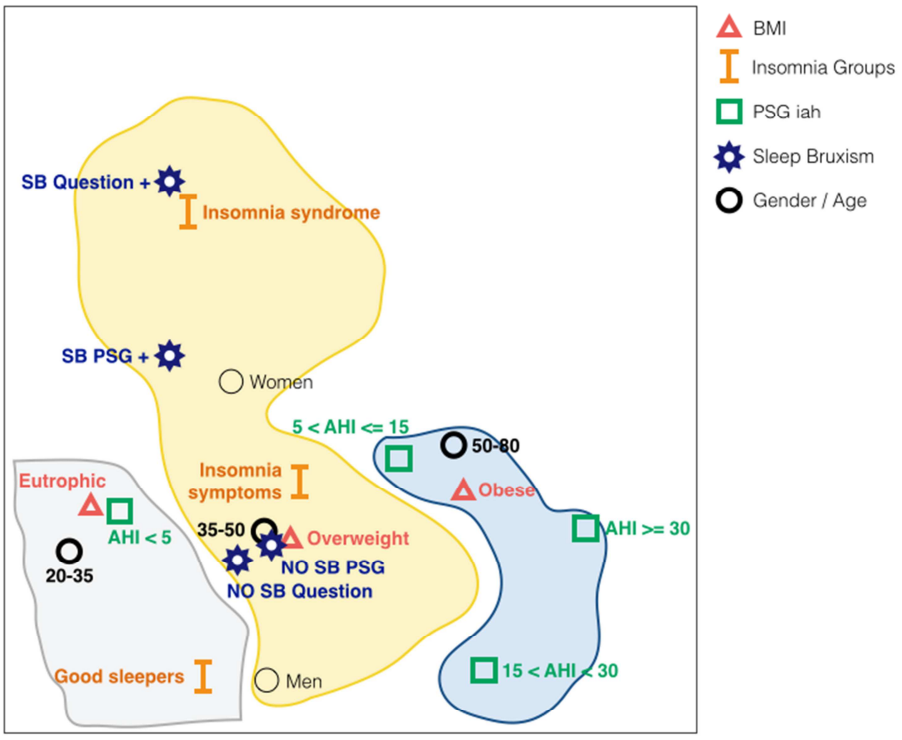
above 15. In summary, in this analysis, the 35- to 50-year-old female group showed overlapping SB and insomnia.

Figure 2– Biplot profile graph of the SB groups and the associated variables using HOMALS.



Cronbach Alpha = .85 - Total Inertia = .50

Legend: BMI – body mass index; AHI (written IAH on figure) – apnea-hypopnea index; SB Question – positive response for tooth grinding, clenching, tapping, or noise during sleep; Current frequency of tooth grinding; PSG SB – presence of ≥2 RMMA episodes per hour; HOMALS: Multiple Correspondence Analysis.

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4. Discussion

In the general population of Sao Paulo, Brazil, insomnia is likely a dominant condition associated with SB in both “possible” and “definitive” identified SB participants. Indeed, it seems that insomnia may be an influential condition in middle-age females, while sleep apnea seems age and gender dependent. Such overlap between co-occurring conditions may influence treatment decisions to achieve best outcomes.

Concomitant insomnia also corresponded to a more accurate prediction of SB as seen in the decision tree analysis. Furthermore, we were able to identify three different clusters: (a) good sleepers aged 20–35 years-old; (b) females aged 35–50 years-old with SB and concomitant insomnia syndrome; and (c) participants aged ≥ 50 years-old with obesity and sleep apnea. Therefore, good sleepers appear to be younger, SB and insomnia are more prevalent in middle age women, and OSA seems to be age dependent, as it increases by age.

The identification of SB is still considered an issue. A better standardization of data collection and analysis methods could help to improve the identification of “possible”, “probable” and “definitive SB” [1,3,13,22,44-46]. In a survey of media professionals, age and gender (female) were reported to be associated with orofacial pain, “possible” bruxism and insomnia [36,37]. The observed cluster of middle-aged female participants with both SB and insomnia is another putative variable of interest that should be included in the development of future studies. Therefore, when investigating risk factors for SB and trying to construct more appropriate SB management algorithms (for diagnosis and treatment) there is a need to control for the presence of comorbidities such as insomnia, sleep apnea and probably mood and orofacial pain. Further studies

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may explore, when more than one condition co-occur in a single individual, the relative dominance of one condition over the other. It remains to be clarified if the coexistence of SB with other sleep disorders such as insomnia and/or OSA, and eventually restless leg syndrome/periodic limb movement or gastroesophageal reflux occurs due to: (i) coincidence of risk factors happening/present in same time due to age, life habits (eg, smoking, caffeine intake, lifestyle, personality, use of medication, etc.); (ii) combination of additional conditions/disorders or symptoms (eg, depression, anxiety, stress) as comorbid event(s); or (iii) naturalistic and independent risk factors or causes resulting in a multimorbidity (ie, co-existence or presence of two or more diseases/conditions observed in same time in a given individual where no single condition holds priority over any of the co-occurring conditions) [47,48].

In the present study we did not further assess if insomnia was related to the occurrence of a given type of RMMA. Phasic RMMA is occurring more frequently during sleep in normo-SB (ie, no comorbidity or multimorbidity as described above), whereas tonic/sustained or mixed events occur much less frequently [49-52]. Further studies challenging the putative association of SB to insomnia and OSA may help to discern the role of phasic, mixed and tonic RMMA events. In a recent report, it was observed that the severity of tension type headache to SB was associated to RMMA mixed event and to sleep arousals. Indeed, such associations explained 34% and 26% of the variability, respectively [53]. The same group observed that the increases in both AHI and oxygen desaturation index were significantly correlated to the number of tonic RMMA events [54]. Therefore, it remains to be explored if such phenotype may explain part of the cause of sleep RMMA in relation to other observed wake or sleep or condition (ie, headache, pain, insomnia, OSA).

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These findings were obtained in a general population, therefore, caution should be taken when interpreting the association between SB by self-report and OSA in clinical populations [18,29,34]. The comorbidities/multimorbidity of SB and OSA may include diabetes, hypertension, and, as observed in the present study, insomnia [18,35,55]. A recent PSG study reported a non-linear association between OSA severity and SB. A positive, low, but significant correlation was found between an AHI below 30 and RMMA frequency (bruxism episode index), and this correlation was negative and non-significant with an AHI of 30 and over [18]. For a conclusive classification of normo-bruxism and patho-bruxism variables such as age, gender, sleep pattern, and medical or mood disorders and associated symptoms need to be taken into consideration in addition to the AHI or RMMA index. [2,3,56]. Moreover, sleep laboratory PSG is not the optimal method to capture time-to-time fluctuations in the RMMA index as recording more than one consecutive night may not be feasible, and the first night effect also has to be considered. Ambulatory home recording may better capture SB motor activity with a few nights' recording in a natural setting [21,44,54,57].

The identification of risk factors is essential for the early detection and successful management of disease, especially in life-threatening conditions such as OSA. Indeed, being male and overweight or obese are considered dominant risk factors for OSA. Females have a number of risk factors that increase with age, including increased levels of inflammatory mediators, menopause, pain, and fatigue [58-61]. In the present study, the correspondence analysis showed a predominance of SB in middle-aged females with insomnia syndrome. Although this result should be interpreted cautiously due to the nature of our population, it highlights the importance of taking into account other comorbidities such as insomnia when identifying SB. This may potentially help clinicians

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to identify sleep disorders through the recognition of phenotypes or patient characteristics. For instance, our findings may be especially relevant for the clinical management of female patients with SB, insomnia, and other conditions such as temporomandibular disorders (TMD), which are frequently associated with both sleep disorders [13,36,37,62-69]. Hence, the potential association of SB, insomnia and OSA in particular phenotypes such as the ones identified in this study, may help dentists and physicians to better assess and manage concomitant conditions, emphasizing the importance of interdisciplinary collaboration for the delivery of successful personalized medicine. Clearly, more research is needed to confirm these associations and to evaluate their importance in the short- and long-term.

The present study includes some limitations that need to be considered before data interpretation and future extrapolation. First, the study is based on questionnaire data, which is accompanied by some limitations as described above [7,8]. The questionnaire employed in this study is an indirect proxy for SB, with limited validity when its use is not accompanied by an objective assessment (ie, clinical examination) [3,19]. An ideal valid and easy to use tool or method to assess SB in research and clinical settings has yet to be developed [22]. Second, PSG recording was conducted for only one night, and in a sleep laboratory; as raised above, then the results may not be representative of typical sleep patterns in the home environment, and cannot account for time-to-time variability in RMMA [44,57,70]. Advances in high-quality home recording devices offer a promising direction [44,45,71]. Third, the low incidence of severe OSA events (ie, AHI>30) in the present study means that caution should be used when considering a possible association between OSA severity and SB; age seems to be the critical factor in OSA, and OSA specific phenotypes for comorbid SB still have to be demonstrated. Fourth, the CHAID analysis compared the data of 1042 participants with

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a niche of only 127 SB cases, which also requires caution. Finally, the presence of an association identified through correspondence analysis does not explain causality. Therefore, the observed association between insomnia and SB should be further investigated, ideally in diverse geographic locations and using home sleep testing to collect real-life exacerbating factors. Moreover, future studies collecting self-report and jaw motor activity data during wakefulness or the presence of oral habits [72] would be of special interest. Research in patients with associated conditions, such as TMD females, who also present persistent elevated muscle tone and increased hyperarousal during sleep [66,68], may be particularly useful in investigating these associations.

5. Conclusions

Concomitant insomnia seems to be consistently associated with SB, based on self-report and PSG, in the general population, especially in middle-aged women. Moreover, SB and sleep apnea associations seem to be age and gender dependent. Future studies assessing SB risk factors, prevalence, mechanisms, or management, should consider the influence of concomitant sleep disorders such as insomnia and OSA.

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Figure 1 – Decision tree analysis (CHAID Algorithm*) to identify the variables that best contribute to detecting sleep bruxism (age, gender, body mass index [BMI], insomnia syndrome). Model A is based on self-report sleep of bruxism (SB) assessments ("possible" SB), while model B is based on self-report and polysomnographic SB assessments ("definitive" SB), including the apnea-hypopnea index.

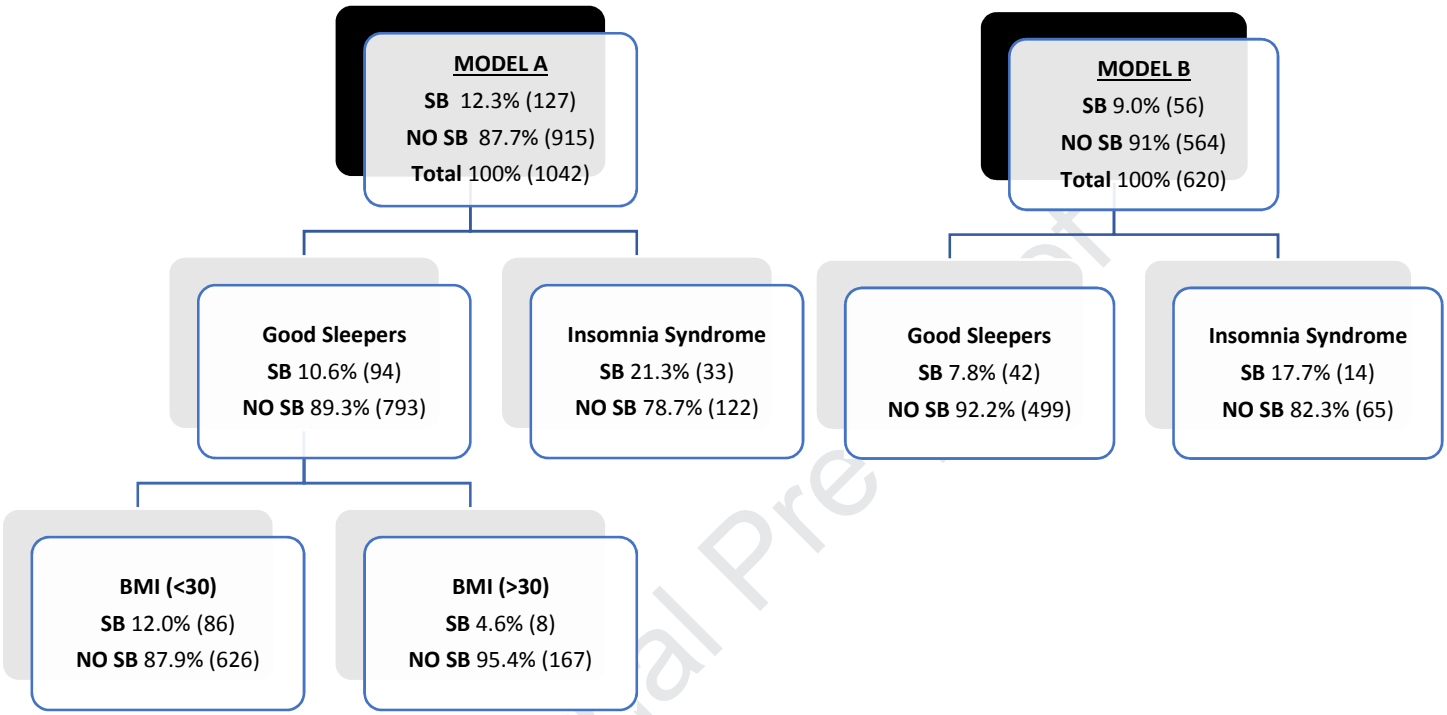

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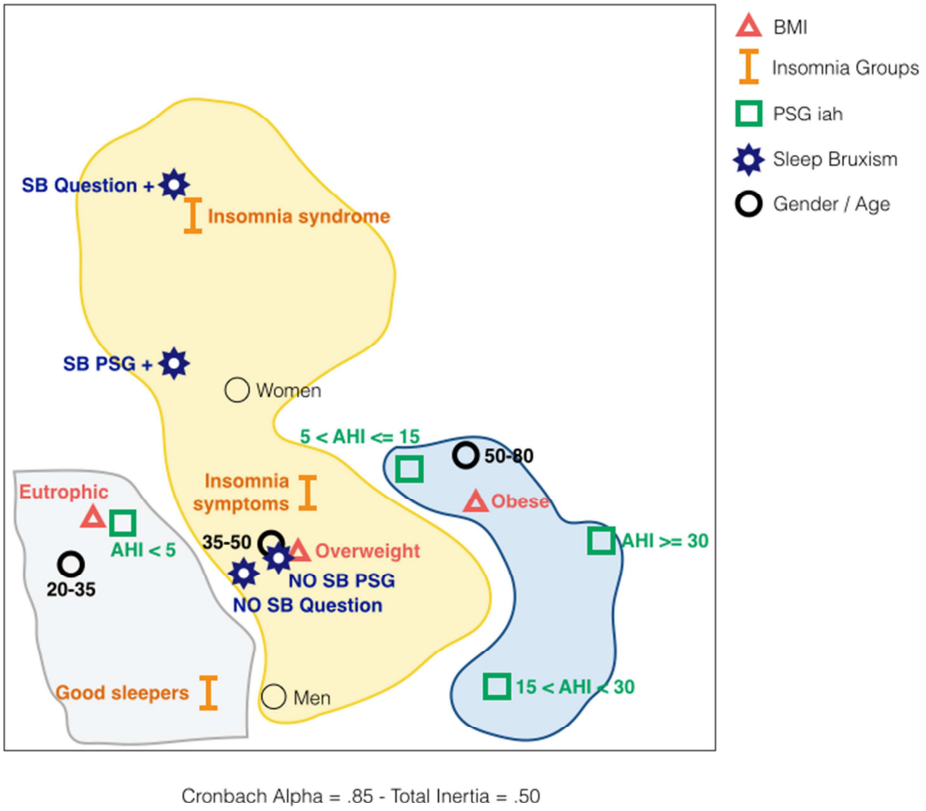
Figure 2– Biplot profile graph of the SB groups and the associated variables using HOMALS.



Legend: BMI – body mass index; AHI (written IAH on figure) – apnea-hypopnea index; SB Question – positive response for tooth grinding, clenching, tapping, or noise during sleep; Current frequency of tooth grinding; PSG SB – presence of ≥2 RMMA episodes per hour; HOMALS: Multiple Correspondence Analysis.

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Highlights

- Insomnia seems to be associated with sleep bruxism in general population.
- The association between insomnia and sleep bruxism is stronger in middle-aged women.
- Sleep bruxism and sleep apnea associations seem to be age and gender dependent.

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